



M Ű E G Y E T E M 1 7 8 2
BUDAPEST UNIVERSITY OF TECHNOLOGY AND ECONOMICS

Neutron and Gamma Transport Calculation Methods
BMETE80NE21

Homework Project
Modelling of the BME Training Reactor using MCNP

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Contents

1. Task Description	3
2. Geometry	4
2.1 Fuel Rod	5
2.2 Cells	5
2.3 Container & Importance Layers	7
3. Materials	8
4. Simulation	9
4.1 Setup	9
4.2 Conversion	10
4.3 Results	12
Appendix	14

1. Task Description

The goal of this homework project is the modelling of the BME Training Reactor using MCNP, following the benchmark description of the facility [1]. A simplified model of the reactor core is built, submerged in a large body of water, and a concrete container is defined around that water as a shield. The quantity of interest is the neutron dose rate on the outer surface of this concrete container.

The training reactor is characterised by its nominal thermal power of 100 kW. This power fixes the absolute neutron source strength of the core, which is used to scale the simulated, per source neutron, flux into an absolute dose rate at full power. The model is solved as an MCNP criticality calculation in neutron transport mode.

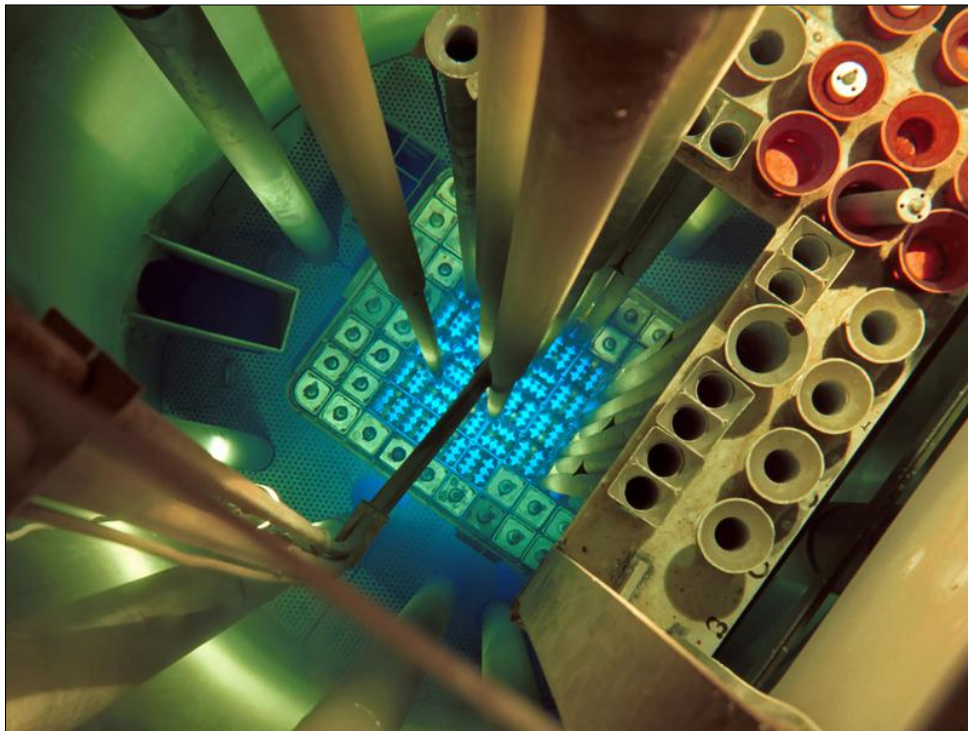


Figure 1: Reactor core of the Training Reactor as seen from above. [1]

2. Geometry

The geometry of the model was built using MCNP [2] and inspected with the Visual Editor [3]. An overview of the reactor core is shown in figure 2, where the core is an 8×9 lattice of cells, addressed by the column letters A to H and the row numbers 1 to 9. The per cell aluminium casings are not drawn so that the cell contents stay visible, but these are shown later in the cell geometry descriptions. Each grid position holds a fuel cell, a graphite control cell or a water filled empty cell.

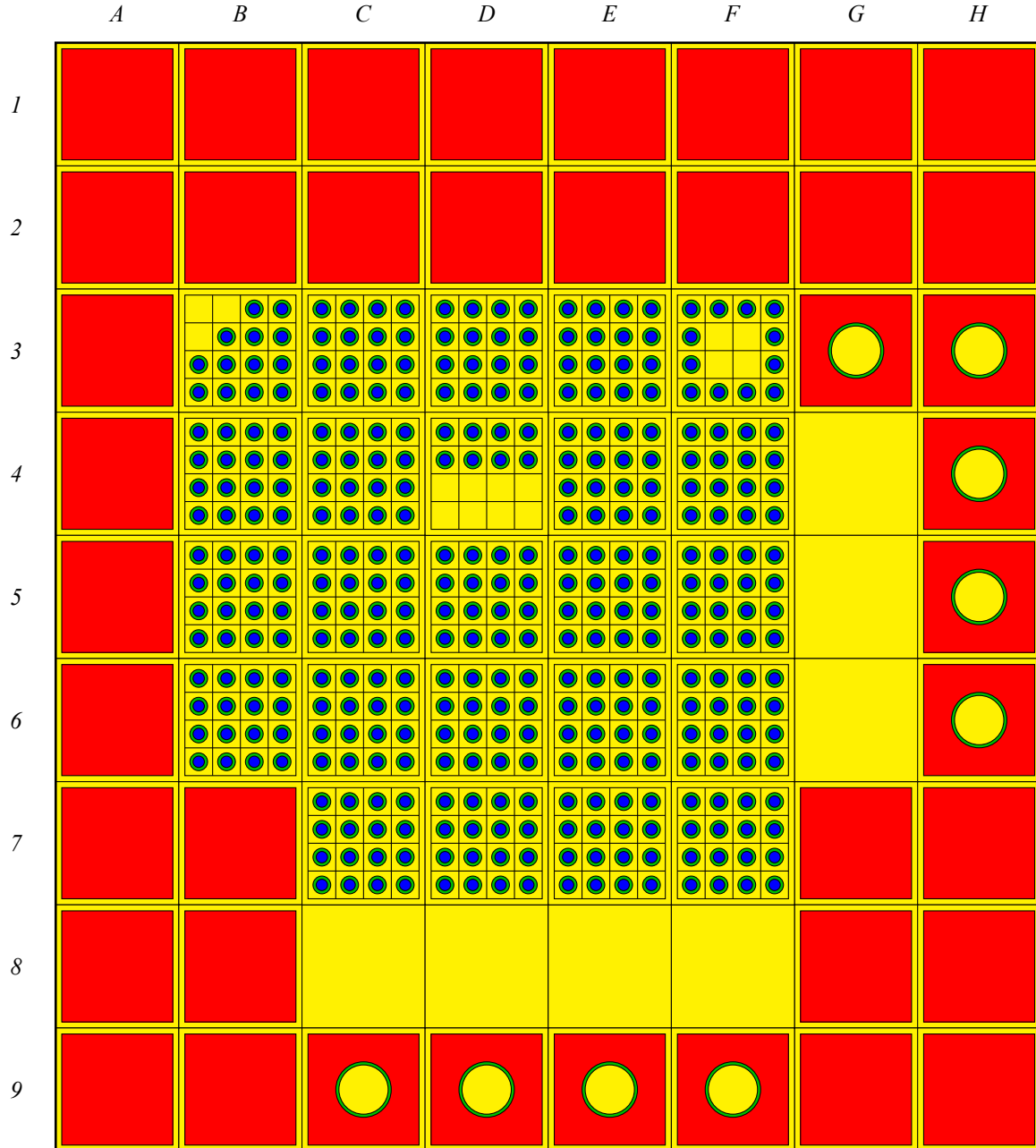


Figure 2: Reactor core geometry showing the lattice of cells. The colours represent materials, blue is UO_2 fuel, green is aluminum, yellow is water and red is graphite.

2.1 Fuel Rod

The fuel rod is modelled as two coaxial cylinders, a UO_2 fuel meat enclosed in an aluminium cladding. Its principal dimensions are given in the blueprint of figure 3.

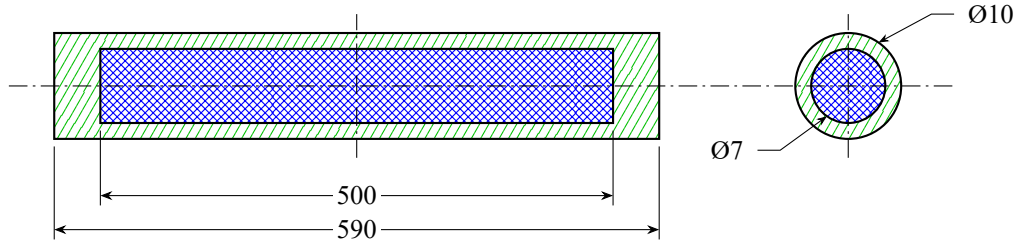


Figure 3: Dimensions of a fuel rod in mm.

2.2 Cells

The fuel rods are fitted into a 4×4 lattice inside an aluminium casing, which together form a fuel cell. Its top view dimensions are shown in figure 4.

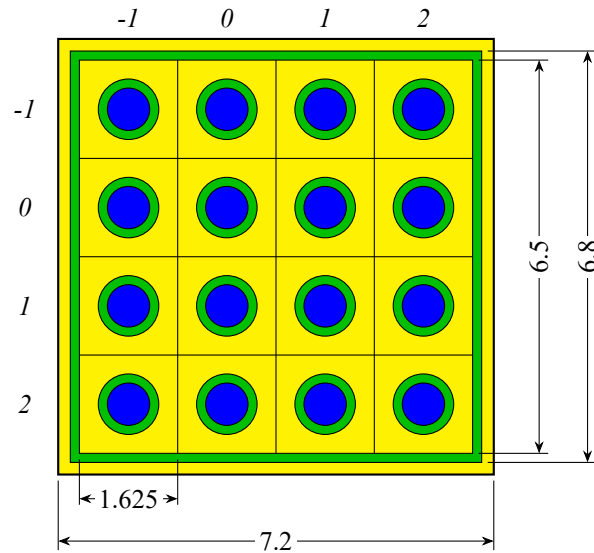


Figure 4: Top view of a fuel cell. A rod is addressed with the lattice coordinates, columns i and rows j . The dimensions are in cm.

The height of each cell, and therefore the whole reactor core is 60 cm.

Besides the fully loaded cell, three partially loaded fuel cell variants occur in the core, a half-filled, a middle-empty and a corner-empty arrangement, shown in figure 5. In each variant the omitted lattice positions are left as water.

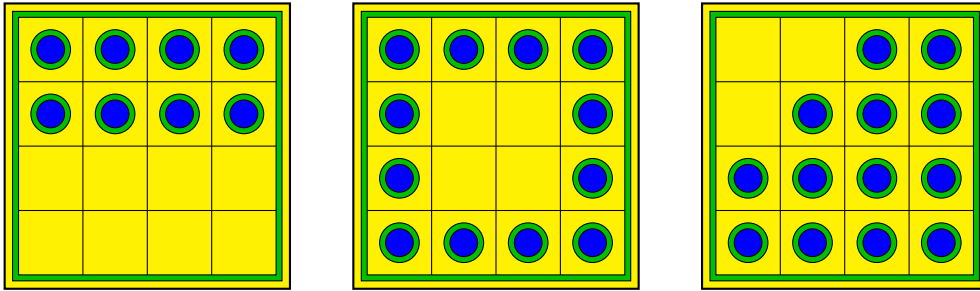


Figure 5: The three partially loaded fuel-cell variants.

The control cells use the same outer aluminium casing but are filled with graphite instead of fuel rods. Some are fully filled, while the holed control cells contain a central, aluminium-lined water channel. Both types are shown in figure 6.

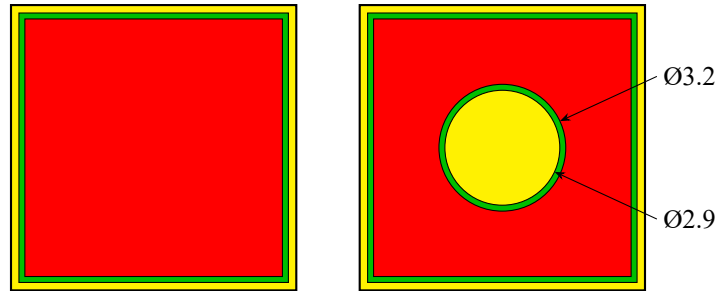


Figure 6: Top view of the two control-cell types.

In the MCNP input file each cell type is a separate universe, empty cells in the reactor core are filled with water.

2.3 Container & Importance Layers

The whole core is submerged in water, which has a diameter of 140 cm, and a cylindrical concrete container surrounds that, the width of the container wall is 50 cm and the height is 200 cm. Both cylinders are divided into 10 cm wide concentric importance layers, with the neutron importance doubled from one shell to the next. Their top view layout is shown in figure 7.

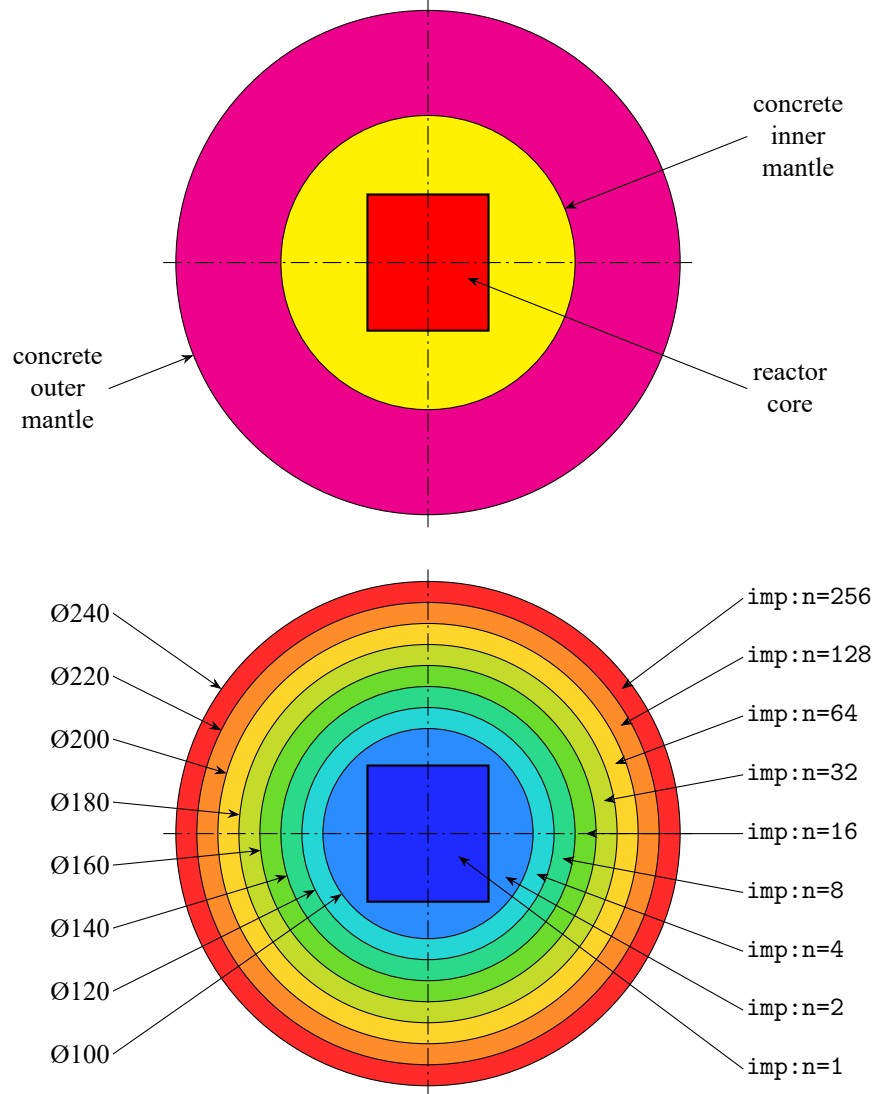


Figure 7: Importance layers around the core, coloured by material (top) and by neutron importance (bottom). The diameters are in cm.

The diameter of the innermost layer is set so that it fully encloses the reactor core. The effective diameter of the core is the diagonal of the 8×9 lattice with a cell pitch of $p = 7.2$ cm, given by the following equation.

$$D_{\text{core}} = \sqrt{(8p)^2 + (9p)^2} = p\sqrt{8^2 + 9^2} = 7.2\sqrt{145} \text{ cm} \approx 86.7 \text{ cm}. \quad (1)$$

The innermost layer is given a diameter of 100 cm, the smallest round value that still contains the whole core.

3. Materials

The materials assigned to the cells and shells of the model are summarised in table 1.

Table 1: Materials of the model, their densities and compositions.

Material	Density [g/cm ³]	Component	Weight fraction
UO ₂ fuel	5.3	²³⁵ U	0.1
		²³⁸ U	0.9
		O	2
Water	1.0	H	2
		O	1
Aluminium	2.7	Al	1
Graphite	1.7	C	1
Concrete	2.4	H	0.01
		C	0.001
		O	0.529107
		Na	0.016
		Mg	0.002
		Al	0.033872
		Si	0.337021
		K	0.013
		Ca	0.044
		Fe	0.014

In MCNP each nuclide is identified by a ZAID of the form $ZZZAAA.nnX$, whose numerical part is built from the atomic number Z and the mass number A as

$$ZZZAAA = 1000 Z + A. \quad (2)$$

For example 92235 stands for $Z = 92$ and $A = 235$, that is U-235, while 8016 stands for oxygen-16. The suffix $.nnX$ names the cross-section evaluation, here the continuous-energy libraries $.50c$, $.51c$ and $.55c$.

The value following each ZAID is the fraction of that nuclide in the material. A positive value is an atom fraction and a negative value is a weight fraction.

4. Simulation

The model is run as an MCNP KCODE criticality (k_{eff}) calculation in neutron transport mode (mode n). The neutron flux leaking through the outer concrete mantle is tallied, folded with the ICRP-116 fluence-to-effective-dose coefficients [4] and scaled to the nominal 100 kW power to finally obtain an absolute dose rate.

4.1 Setup

The criticality source is controlled by the KCODE card and seeded by a single KSRC point. The four KCODE parameters are listed in table 2.

kcode	2000	1.0	20	220
ksrc	-1.1625	0.8125	0	

Table 2: Parameters of the KCODE criticality card.

Parameter	Description	Value
NSRCK	Source histories per cycle	2000
RKK	Initial guess for k_{eff}	1.0
IKZ	Inactive (skipped) cycles	20
KCT	Total number of cycles	220

The number of neutron histories that contribute to the tallies is the number of source neutrons per cycle multiplied by the number of active cycles, which is the total number of cycles minus the inactive ones.

$$N = \text{NSRCK} \times (\text{KCT} - \text{IKZ}) = 2000 \times (220 - 20) = 4.0 \times 10^5. \quad (3)$$

The first $\text{IKZ} = 20$ cycles are inactive and discarded while the fission source converges, so of the 4.4×10^5 histories run in total, only these 4.0×10^5 active histories contribute to the tallies.

The KSRC seed is placed at the centre of a single fuel rod, the one at lattice position 2:0 (column $i = 2$, row $j = 0$) inside the fuel cell at core position D5. From this single point the fission source spreads over all fuel cells during the inactive cycles.

The dose rate is scored on the outer surface of the concrete container, the $r = 120$ cm cylinder, with a neutron surface-flux tally (F2). The fluence is split into three energy groups by an E2 card with edges at 0.5×10^{-6} , 0.5 and 20 MeV, implementing the conventional thermal, epithermal and fast classification:

$$\begin{aligned} \text{thermal: } & E \leq 0.5 \text{ eV,} \\ \text{epithermal: } & 0.5 \text{ eV} < E \leq 0.5 \text{ MeV,} \\ \text{fast: } & 0.5 \text{ MeV} < E \leq 20 \text{ MeV.} \end{aligned} \quad (4)$$

The 0.5 eV edge is the cadmium cut-off, the upper limit of the near-Maxwellian thermal group whose flux peaks near $kT \approx 0.025$ eV, below which cadmium absorbs strongly and above which it is essentially transparent. The 0.5 MeV edge is the conventional lower limit of the fast fission-spectrum group, and 20 MeV is the upper energy limit of the continuous-energy cross-section libraries.

4.2 Conversion

The F2 tally scores the flux per source (fission) neutron, so turning it into an absolute dose rate at the nominal power requires the neutron source strength of the core, the ICRP fluence-to-dose coefficients [4] and a conversion of the units. These are supplied by the FM2 and DE2/DF2 cards and are derived below.

The source strength S is the fission rate $\dot{n}_{\text{fis}} = P/E_f$ times the mean number of neutrons released per fission $\bar{\nu}$:

$$\dot{n}_{\text{fis}} = \frac{P}{E_f}, \quad S = \bar{\nu} \dot{n}_{\text{fis}} = \frac{\bar{\nu} P}{E_f}. \quad (5)$$

With the thermal power $P = 100 \text{ kW} = 1.0 \times 10^5 \text{ J/s}$, the standard mean yield of ^{235}U thermal fission $\bar{\nu} = 2.43$, and an effective energy release of $E_f = 186 \text{ MeV}$ per fission, the fission rate and source strength take the following values.

$$\begin{aligned} \dot{n}_{\text{fis}} &= \frac{1.0 \times 10^5 \text{ J/s}}{186 \text{ MeV} \times 1.602 \times 10^{-13} \text{ J/MeV}} = 3.36 \times 10^{15} \text{ fissions/s}, \\ S &= 2.43 \times 3.36 \times 10^{15} \text{ fissions/s} = 8.15 \times 10^{15} \text{ n/s}. \end{aligned} \quad (6)$$

Folding the flux with the ICRP-116 coefficients collected in table 3 gives a dose in pSv per source neutron, and multiplying by S yields pSv/s, whereas the result is wanted in $\mu\text{Sv/h}$, therefore the following unit conversion is applied to the factor.

$$\frac{\text{pSv}}{\text{s}} \rightarrow \frac{\mu\text{Sv}}{\text{h}} : \quad 3.6 \times 10^3 \frac{\text{s}}{\text{h}} \times 10^{-6} \frac{\mu\text{Sv}}{\text{pSv}} = 3.6 \times 10^{-3}. \quad (7)$$

The source strength and this factor are combined into the single tally multiplier.

$$\text{FM2} = S \times 3.6 \times 10^{-3} = 8.15 \times 10^{15} \times 3.6 \times 10^{-3} = 2.934 \times 10^{13}. \quad (8)$$

The energy-dependent response is the ICRP-116 [4] neutron fluence-to-effective-dose conversion for the antero-posterior (AP) irradiation geometry, in which the radiation reaches the body from front to back. It is entered through the DE2/DF2 cards as the 48 energy and coefficient pairs listed in table 3 and plotted in figure 8.

Table 3: Neutron fluence-to-effective-dose conversion coefficients (ICRP-116, AP geometry).

E [MeV]	h [pSv cm ²]	E [MeV]	h [pSv cm ²]	E [MeV]	h [pSv cm ²]
1.0×10^{-9}	3.09	2.0×10^{-3}	7.61	1.5	365
1.0×10^{-8}	3.55	5.0×10^{-3}	7.97	2.0	407
2.5×10^{-8}	4.00	1.0×10^{-2}	9.11	3.0	458
1.0×10^{-7}	5.20	2.0×10^{-2}	12.2	4.0	483
2.0×10^{-7}	5.87	3.0×10^{-2}	15.7	5.0	494
5.0×10^{-7}	6.59	5.0×10^{-2}	23.0	6.0	498
1.0×10^{-6}	7.03	7.0×10^{-2}	30.6	7.0	499
2.0×10^{-6}	7.39	1.0×10^{-1}	41.9	8.0	499
5.0×10^{-6}	7.71	1.5×10^{-1}	60.6	9.0	500
1.0×10^{-5}	7.82	2.0×10^{-1}	78.8	10	500
2.0×10^{-5}	7.84	3.0×10^{-1}	114	12	499
5.0×10^{-5}	7.82	5.0×10^{-1}	177	14	495
1.0×10^{-4}	7.79	7.0×10^{-1}	232	15	493
2.0×10^{-4}	7.73	9.0×10^{-1}	279	16	490
5.0×10^{-4}	7.54	1.0	301	18	484
1.0×10^{-3}	7.54	1.2	330	20	477

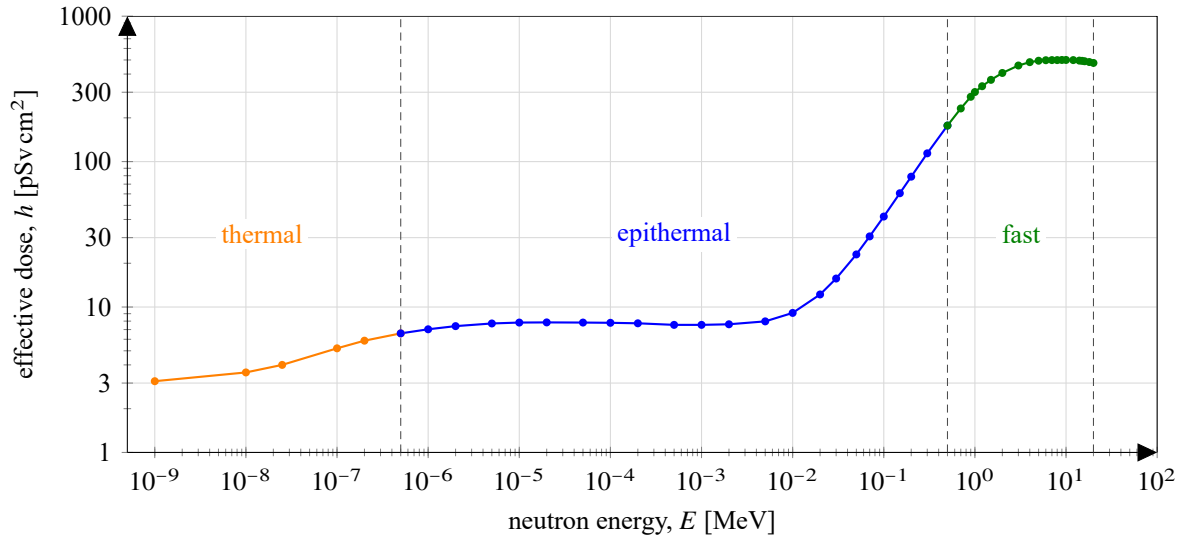


Figure 8: The conversion coefficients of table 3 plotted on logarithmic axes.

4.3 Results

The dose rates obtained over the active cycles are collected in table 4, split into the thermal, epithermal and fast groups.

Table 4: Neutron dose rate on the outer concrete surface ($r = 120$ cm) at 100 kW reactor power, by energy group.

Energy group	Energy range	Dose rate [$\mu\text{Sv/h}$]	Rel. error
Thermal	≤ 0.5 eV	1.114×10^4	0.071
Epithermal	0.5 eV – 0.5 MeV	2.244×10^4	0.126
Fast	0.5 – 20 MeV	4.226×10^5	0.126
Total	—	4.562×10^5	0.121

The total neutron dose rate on the outer mantle is about $4.56 \times 10^5 \mu\text{Sv/h}$ (roughly 456 mSv/h), and it is dominated by the fast group, which carries about 93% of the total. The criticality calculation itself converged to a multiplication factor of $k_{\text{eff}} = 1.04682 \pm 0.00119$.

The relevant parts of the output file are the converged k_{eff} , the energy-binned tally and the statistical-check summary:

```

the final estimated combined keff = 1.04682
with an estimated standard deviation of 0.00119

tally 2  particle flux averaged over a surface (neutron)  nps = 440335
  energy      dose [uSv/h]  rel err
  5.0000E-07    1.11366E+04   0.0706
  5.0000E-01    2.24432E+04   0.1260
  2.0000E+01    4.22609E+05   0.1258
  total        4.56189E+05   0.1205

warning. the tally in the tally fluctuation chart bin did not
pass 3 of the 10 statistical checks.
```

The tally converged only to a relative error of about 12% and failed 3 of the 10 MCNP statistical checks, so the value is indicative rather than fully converged. A longer run with more active cycles, or a stronger variance reduction, would be needed for a reliable estimate. As an example, the KCODE card below raises the source histories per cycle to 20000 and the active cycles to 500, giving 10^7 active histories.

```
kcode 20000 1.0 20 520
```

References

- [1] Gergely Klujber and András Horváth. *Benchmark Description of the BME Training Reactor*. Technical Report BME-NTI-854/2018. Version 3.0. Budapest, Hungary: Budapest University of Technology and Economics, Institute of Nuclear Techniques, Oct. 2020.
- [2] J. K. Shultis and R. E. Faw. *An MCNP Primer*. Department of Mechanical and Nuclear Engineering, Kansas State University. Manhattan, KS 66506, USA, 2006.
- [3] A. L. Schwarz, R. A. Schwarz, and L. L. Carter. *MCNP/MCNPX Visual Editor Computer Code Manual*. For Vised Version 24E, compiled with MCNPX version 2.7. Feb. 2011. URL: <http://www.mcnpvised.com>.
- [4] ICRP. “Conversion Coefficients for Radiological Protection Quantities for External Radiation Exposures”. In: *Annals of the ICRP* 40.2–5 (2010). ICRP Publication 116. ISSN: 0146-6453.

Appendix

HW_QTU3QF.txt MCNP input file:

```
1 HW_QTU3QF
2 c
3 c CELLS
4 c
5 c fuel rod
6 1 1 -5.3 -1 -4 5 imp:n=1 u=1 $ meat
7 2 3 -2.7 -2 -6 7 #1 imp:n=1 u=1 $ cladding
8 3 2 -1.0 #1 #2 imp:n=1 u=1 $ water gap
9 c
10 c fuel cell: filled
11 10 2 -1.0 -11 12 14 -13 -55 56 lat=1 fill=-1:2 -1:2 0:0
12 1 1 1 1
13 1 1 1 1
14 1 1 1 1
15 1 1 1 1 imp:n=1 u=2 $ fuel grid
16 11 0 -21 22 -23 24 -55 56 fill=2 imp:n=1 u=3 $ cell inner box
17 12 3 -2.7 -31 32 -33 34 -55 56 #11 imp:n=1 u=3 $ casing
18 13 2 -1.0 #(-31 32 -33 34 -55 56) imp:n=1 u=3 $ water gap
19 c
20 c fuel cell: half-filled
21 20 2 -1.0 -11 12 14 -13 -55 56 lat=1 fill=-1:2 -1:2 0:0
22 1 1 1 1
23 1 1 1 1
24 4 4 4 4
25 4 4 4 4 imp:n=1 u=4 $ fuel grid
26 21 0 -21 22 -23 24 -55 56 fill=4 imp:n=1 u=5 $ cell inner box
27 22 3 -2.7 -31 32 -33 34 -55 56 #21 imp:n=1 u=5 $ casing
28 23 2 -1.0 #(-31 32 -33 34 -55 56) imp:n=1 u=5 $ water gap
29 c
30 c fuel cell: middle-empty
31 30 2 -1.0 -11 12 14 -13 -55 56 lat=1 fill=-1:2 -1:2 0:0
32 1 1 1 1
33 1 6 6 1
34 1 6 6 1
35 1 1 1 1 imp:n=1 u=6 $ fuel grid
36 31 0 -21 22 -23 24 -55 56 fill=6 imp:n=1 u=7 $ cell inner box
37 32 3 -2.7 -31 32 -33 34 -55 56 #31 imp:n=1 u=7 $ casing
38 33 2 -1.0 #(-31 32 -33 34 -55 56) imp:n=1 u=7 $ water gap
39 c
40 c fuel cell: corner-empty
41 40 2 -1.0 -11 12 14 -13 -55 56 lat=1 fill=-1:2 -1:2 0:0
42 8 8 1 1
43 8 1 1 1
44 1 1 1 1
45 1 1 1 1 imp:n=1 u=8 $ fuel grid
46 41 0 -21 22 -23 24 -55 56 fill=8 imp:n=1 u=9 $ cell inner box
47 42 3 -2.7 -31 32 -33 34 -55 56 #41 imp:n=1 u=9 $ casing
48 43 2 -1.0 #(-31 32 -33 34 -55 56) imp:n=1 u=9 $ water gap
49 c
50 c control cell: full
51 50 4 -1.7 -21 22 -23 24 -55 56 imp:n=1 u=10 $ reflector body
52 51 3 -2.7 -31 32 -33 34 -55 56 #50 imp:n=1 u=10 $ casing
53 52 2 -1.0 #(-31 32 -33 34 -55 56) imp:n=1 u=10 $ water gap
54 c
55 c control cell: holed
56 60 2 -1.0 -8 -55 56 imp:n=1 u=11 $ central hole
57 61 3 -2.7 -9 8 -55 56 imp:n=1 u=11 $ hole casing
58 62 4 -1.7 -21 22 -23 24 -55 56 #60 #61 imp:n=1 u=11 $ reflector body
59 63 3 -2.7 -31 32 -33 34 -55 56 #60 #61 #62 imp:n=1 u=11 $ casing
60 64 2 -1.0 #(-31 32 -33 34 -55 56) imp:n=1 u=11 $ water gap
61 c
```

```

62 c core grid
63 100 2 -1.0 -41 42 44 -43 -55 56 lat=1 fill=-3:4 -4:4 0:0
64 10 10 10 10 10 10 10 10 $ row 1
65 10 10 10 10 10 10 10 10 $ row 2
66 10 9 3 3 3 7 11 11 $ row 3
67 10 3 3 5 3 3 20 11 $ row 4
68 10 3 3 3 3 3 20 11 $ row 5
69 10 3 3 3 3 3 20 11 $ row 6
70 10 10 3 3 3 3 10 10 $ row 7
71 10 10 20 20 20 20 10 10 $ row 8
72 10 10 11 11 11 11 10 10 imp:n=1 u=20 $ row 9
73 c A B C D E F G H
74 c
75 c importance layers
76 200 0 -51 52 -53 54 -55 56 fill=20 imp:n=1 trcl=1 $ center the core
77 201 2 -1.0 #200 -61 -71 72 imp:n=2 $ water r 0-50
78 202 2 -1.0 -62 61 -71 72 imp:n=4 $ water r 50-60
79 203 2 -1.0 -63 62 -71 72 imp:n=8 $ water r 60-70
80 204 5 -2.4 -64 63 -71 72 imp:n=16 $ concrete r 70-80
81 205 5 -2.4 -65 64 -71 72 imp:n=32 $ concrete r 80-90
82 206 5 -2.4 -66 65 -71 72 imp:n=64 $ concrete r 90-100
83 207 5 -2.4 -67 66 -71 72 imp:n=128 $ concrete r 100-110
84 208 5 -2.4 -68 67 -71 72 imp:n=256 $ concrete r 110-120
85 209 0 #200 #201 #202 #203 #204
86 #205 #206 #207 #208 imp:n=0 $ outside
87
88 c SURFACES
89 c
90 c fuel rod
91 1 cz 0.35 $ meat radius
92 2 cz 0.5 $ clad radius
93 4 pz 25.0 $ meat top
94 5 pz -25.0 $ meat bottom
95 6 pz 29.5 $ clad top
96 7 pz -29.5 $ clad bottom
97 c reflector hole
98 8 c/z 0.8125 -0.8125 1.45 $ hole
99 9 c/z 0.8125 -0.8125 1.60 $ hole casing
100 c fuel rod grid tile
101 11 px 0.8125 $ rod tile +x
102 12 px -0.8125 $ rod tile -x
103 13 py 0.8125 $ rod tile +y
104 14 py -0.8125 $ rod tile -y
105 c cell inner box
106 21 px 4.0625 $ inner box +x
107 22 px -2.4375 $ inner box -x
108 23 py 2.4375 $ inner box +y
109 24 py -4.0625 $ inner box -y
110 c casing
111 31 px 4.2125 $ casing +x
112 32 px -2.5875 $ casing -x
113 33 py 2.5875 $ casing +y
114 34 py -4.2125 $ casing -y
115 c cell grid tile
116 41 px 4.4125 $ cell grid tile +x
117 42 px -2.7875 $ cell grid tile -x
118 43 py 2.7875 $ cell grid tile +y
119 44 py -4.4125 $ cell grid tile -y
120 c reactor core
121 51 px 33.2125 $ core +x
122 52 px -24.3875 $ core -x
123 53 py 31.5875 $ core +y
124 54 py -33.2125 $ core -y
125 55 pz 30.0 $ core top
126 56 pz -30.0 $ core bottom

```

```

127 c importance layers
128 61 cz 50.0 $ water
129 62 cz 60.0
130 63 cz 70.0
131 64 cz 80.0 $ concrete
132 65 cz 90.0
133 66 cz 100.0
134 67 cz 110.0
135 68 cz 120.0 $ container tank mantle
136 c contaier tank
137 71 pz 100.0 $ container tank top
138 72 pz -100.0 $ container tank bottom
139
140 c DATA
141 c
142 c materials
143 m1 92235.50c 0.1 $ U-235
144 92238.50c 0.9 $ U-238
145 8016.50c 2 $ O -> UO2 (rho 5.3)
146 m2 1001.50c 2 $ H
147 8016.50c 1 $ O -> water (rho 1.0)
148 m3 13027.50c 1 $ aluminium (rho 2.7)
149 m4 6012.50c 1 $ graphite (rho 1.7)
150 m5 1001.50c -0.01 $ H
151 6012.50c -0.001 $ C
152 8016.50c -0.529107 $ O
153 11023.51c -0.016 $ Na
154 12000.51c -0.002 $ Mg
155 13027.50c -0.033872 $ Al
156 14000.51c -0.337021 $ Si
157 19000.51c -0.013 $ K
158 20000.51c -0.044 $ Ca
159 26000.55c -0.014 $ Fe -> concrete (rho 2.4)
160 c
161 c transform (to center the reactor core)
162 tr1 -4.4125 0.8125 0
163 c source
164 mode n
165 kcode 2000 1.0 20 220 $ hist/cycle, initial k_eff, inactive, total cycles
166 ksrc -1.1625 0.8125 0 $ D5 cell (1;2) fuel rod lattice element center
167 c
168 c tally (neutron flux -> dose rate)
169 f2:n 68 $ avg neutron flux on container tank mantle
170 fm2 2.934e13 $ 8.15e15 n/s (100 kW) x 3.6e-3 (pSv/s -> uSv/h)
171 e2 0.5e-6 0.5 20 $ energy bins: thermal / epithermal / fast [MeV]
172 c
173 c energy-dependent response function [pSv*cm^2]
174 c (ICRP-116 : Table A.5 : AP geometry)
175 de2 1.0e-9 1.0e-8 2.5e-8 1.0e-7 2.0e-7 5.0e-7 1.0e-6 2.0e-6
176 5.0e-6 1.0e-5 2.0e-5 5.0e-5 1.0e-4 2.0e-4 5.0e-4 1.0e-3
177 2.0e-3 5.0e-3 1.0e-2 2.0e-2 3.0e-2 5.0e-2 7.0e-2 1.0e-1
178 1.5e-1 2.0e-1 3.0e-1 5.0e-1 7.0e-1 9.0e-1 1.0e0 1.2e0
179 1.5e0 2.0e0 3.0e0 4.0e0 5.0e0 6.0e0 7.0e0 8.0e0
180 9.0e0 1.0e1 1.2e1 1.4e1 1.5e1 1.6e1 1.8e1 2.0e1
181 c
182 df2 3.09 3.55 4.00 5.20 5.87 6.59 7.03 7.39
183 7.71 7.82 7.84 7.82 7.79 7.73 7.54 7.54
184 7.61 7.97 9.11 12.2 15.7 23.0 30.6 41.9
185 60.6 78.8 114 177 232 279 301 330
186 365 407 458 483 494 498 499 499
187 500 500 499 495 493 490 484 477
188

```